Why is Evaluation Central to Machine Learning?

In business, a machine learning model is only valuable if it **performs well on real-world data**. Evaluation is how we answer the question:

> *"Can we trust this model to make decisions or predictions in practice?"*

Without proper evaluation:

- You might deploy a model that looks great in development but fails with actual customers.
- Business decisions could be based on unreliable insights.

Two Core Components of Evaluation:

1. **Loss Function (During Training)**
    - **Purpose**: Guides how the model learns by measuring the "error" in predictions.
    - **Used when**: The model is learning from historical data
    - **Example**: If your model predicts this month's revenue as ₹95,000 and the actual is ₹100,000, the **loss** is ₹5,000.
    - **Goal**: Minimize this loss to make better predictions.

      Think of it as: A GPS recalculating your route - the model uses the loss to adjust its internal logic and improve accuracy.

2. **Evaluation Metric (After Training)**
    - **Purpose**: Tells you how good the model is on **new, unseen data**.
    - **Used when**: You want to decide whether to trust the model's results.
    - **Example**: For a fraud detection model, we want to ensure that it flags most of the fraudulent cases.

      Think of it as: The final test after practice - it's your model's performance scorecard.

Why Do We Need Both?

- At the end of the month, you look at their final numbers: revenue, deals closed (**evaluation metric**), to decide if they're ready to go client-facing.

Similarly:

- The **loss function** guides learning.
- The **evaluation metric** assesses final performance.

They **serve different purposes**, and one **can't replace the other**.

Example to Drive It Home

Business Problem: Predict monthly demand for a product.

- The **loss function** helps the model adjust its logic so its demand predictions improve during training (e.g., predicting 960 instead of 800 when the true demand was 1000).
- The **evaluation metric** checks how close the predictions are, **on new data**, to ensure it's ready for business decision making, like inventory or staffing.

Therefore, before we pick the **best model**, we usually try out several with different techniques, settings, or data strategies.

What Does "Best Model" Mean?

> *There's no universal "best" model, only what's best **for your business goal**.*

Before comparing models:

- Be clear whether you care about **accuracy**, **cost reduction**, **catching edge cases**, or **speed**.
- Define what success looks like in your business context.

In the next videos, you'll learn how loss functions and evaluation metrics are used to train and evaluate regression models, and how they help us choose the best one. You'll also see:

metrics.

Power Ahead!